



PYTHON HOMEWORK – ASSIGNMENT 2

FOR LOOP, RANGE & IN OPERATOR (EXTENDED)



Learning Objectives

- Understand for loops
 - Use range() properly
 - Use in operator with lists, strings, and tuples
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◆ Part A: Theory Questions

1. What is a for loop?
 2. Difference between for loop and while loop?
 3. What is range()?
 4. What are the three parameters of range()?
 5. What does the in operator do?
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◆ Part B: Basic for Loop Practice

Problem 1

Print numbers from **1 to 10**.

Problem 2

Print numbers from **10 to 1**.

Problem 3

Print all **odd numbers** from 1 to 50.

Problem 4

Print the square of numbers from **1 to 10**.

Problem 5

Print each character of your name on a new line.

◆ Part C: range() Advanced Practice

Problem 6

Print numbers from **5 to 50**, stepping by **5**.

Problem 7

Print all numbers between **1 and 100** that are divisible by both **3 and 5**.

Problem 8

Print the sum of numbers from **1 to 100** using a for loop.

Problem 9

Print the multiplication table (1–10) for a user-entered number.

◆ Part D: in Operator Practice

Problem 10

Check whether a given number exists in a list.

Problem 11

Check if a character exists in a string entered by the user.

Problem 12

Find how many times a character appears in a string.

◆ Part E: MINI PROJECTS (EXTENDED)

Mini Project 1: Grade Analyzer

Problem Statement:

- Store marks in a list
 - Print each mark
 - Calculate average marks
 - Print pass or fail for each student
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Mini Project 2: Shopping Cart Total

Problem Statement:

- Store item prices in a list
 - Add all prices using a for loop
 - Apply a discount if total > 1000
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Mini Project 3: Word Counter

Problem Statement:

- Take a sentence from the user
 - Count the number of words using a for loop
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Mini Project 4: Pattern Printing

Problem Statement:

Print this pattern using a for loop:

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Mini Project 5: Number Frequency Counter

Problem Statement:

- Store numbers in a list
- Count how many times each number appears